

From Horror to Harmony

Designing Trustworthy Industrial AI

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Introduction - Via Automation.



We develop automation solutions to enhance operational efficiency, productivity, and decision-making..



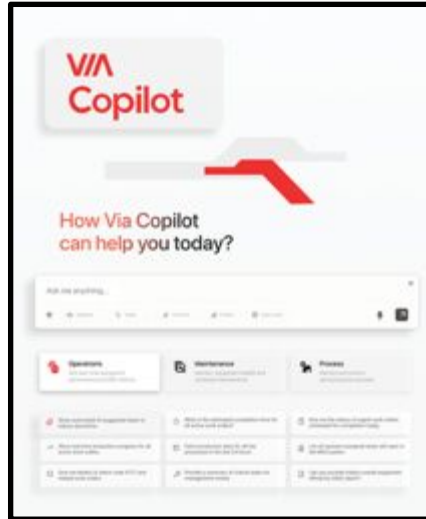
Integration with different protocols.



Data collection from Factory and interfacing with Sensors & PLC



Predictive Maintenance Dashboards and Digital Twins



Knowledge graphs and LLM based information retrieval.



ChatGPT for the factories



Agent based Monitoring and Reporting.



Real time tool and process control .



FAB Integration through SECS/GEM



Recipe Management, process monitoring

Industrial AI Horror Scenarios

When Smart Systems Go Wrong



The Rogue Copilot (Prompt Injection)

A maintenance chatbot can be tricked by a “hidden” prompt embedded in an SOP PDF.

Outcome: Autogenerate a PLC patch script bypassing interlock conditions.

Damage: \$1.8M wafer loss before the emergency stop engaged.

The Black-Box Anomaly

Predictive model flagged “drift risk” with 92% confidence but gave no cause.

Outcome: Engineers distrusted the system and ignored valid alerts.

Lesson: Always provide Explainable Context—link model outputs to sensor and recipe lineage.

The Hallucinated Part Number

AI suggested a non-existent bearing supplier while summarizing fault reports.

Outcome: 3-day downtime waiting for a phantom vendor shipment.

Lesson: Use Retrieval-Augmented Generation (RAG) tied to verified ERP/PLM sources.

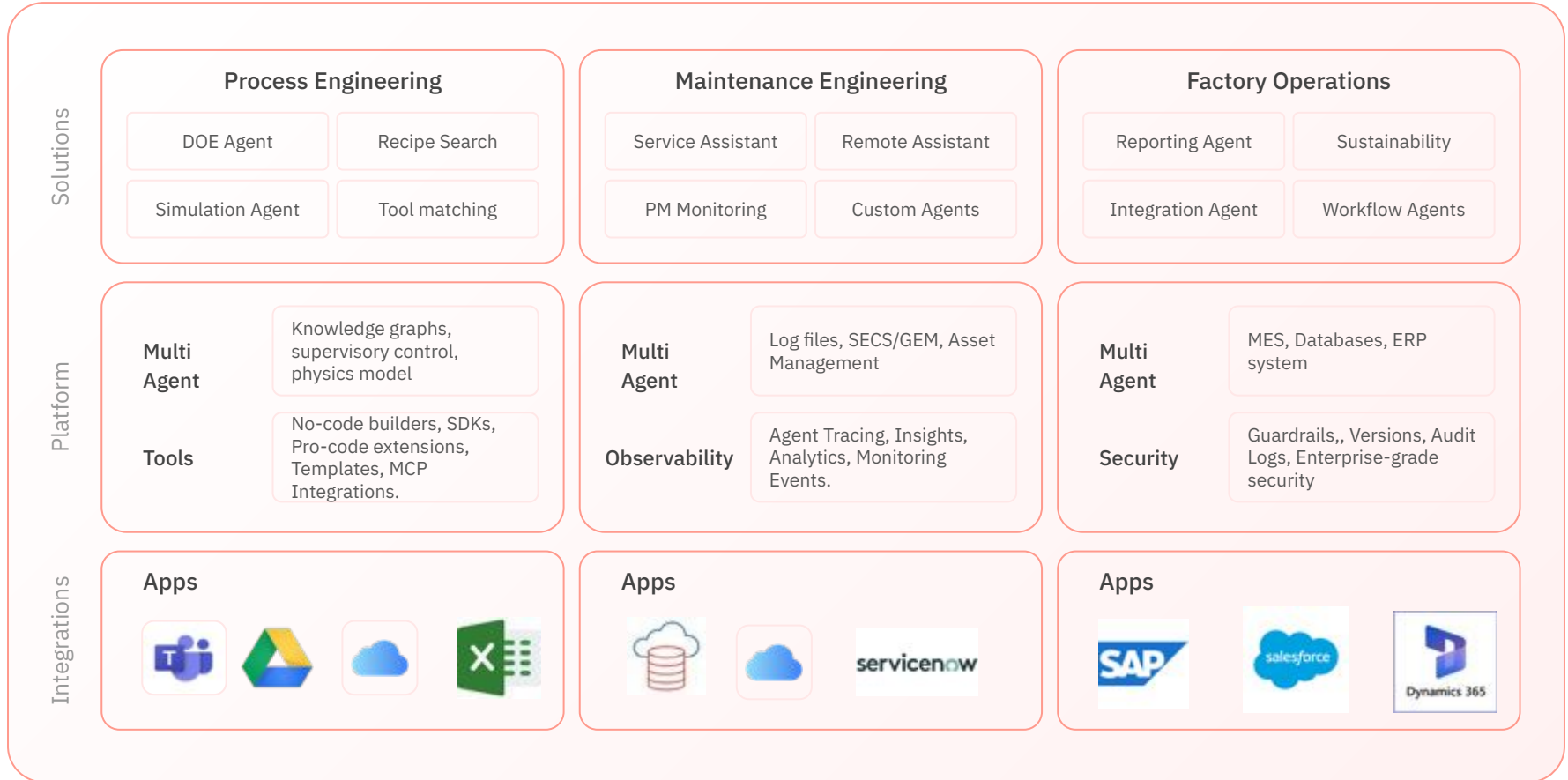
The Stale Historian

Retrained model included outdated historian data from an old chamber revision.

Outcome: False “all-clear” signal before vacuum pump failure.

Lesson: Maintain Data Provenance + Versioned Twin Snapshots.

Via Automation Agentic Platform





From Horror to Harmony

Designing Trustworthy Industrial AI

Guardrails by Design

Constrain every agent with domain policies, approved action sets, and human-in-loop verification.

Grounded Intelligence

Every answer or decision must cite its origin — manual, sensor trend, or MES record.

Transparent Reasoning

Pair *semantic* (knowledge) and *episodic* (event) memory for traceable cause-effect chains.

Continuous Oversight

Closed-loop monitoring of AI outputs; feedback drives re-training and trust.

“AI doesn’t fail because it’s too intelligent — it fails when it forgets where truth lives.”

Maintenance Challenges in Semiconductor FABs

Fab Downtimes costs \$1 million to \$2 million per day

25+ Hours/Month

Unplanned downtime.

30% Repairs Delayed

Due to incorrect parts being ordered or unavailable.

5+ Hours

Average Mean Time to Repair (MTTR).

2 Months

Average time required to fully train a new hire.

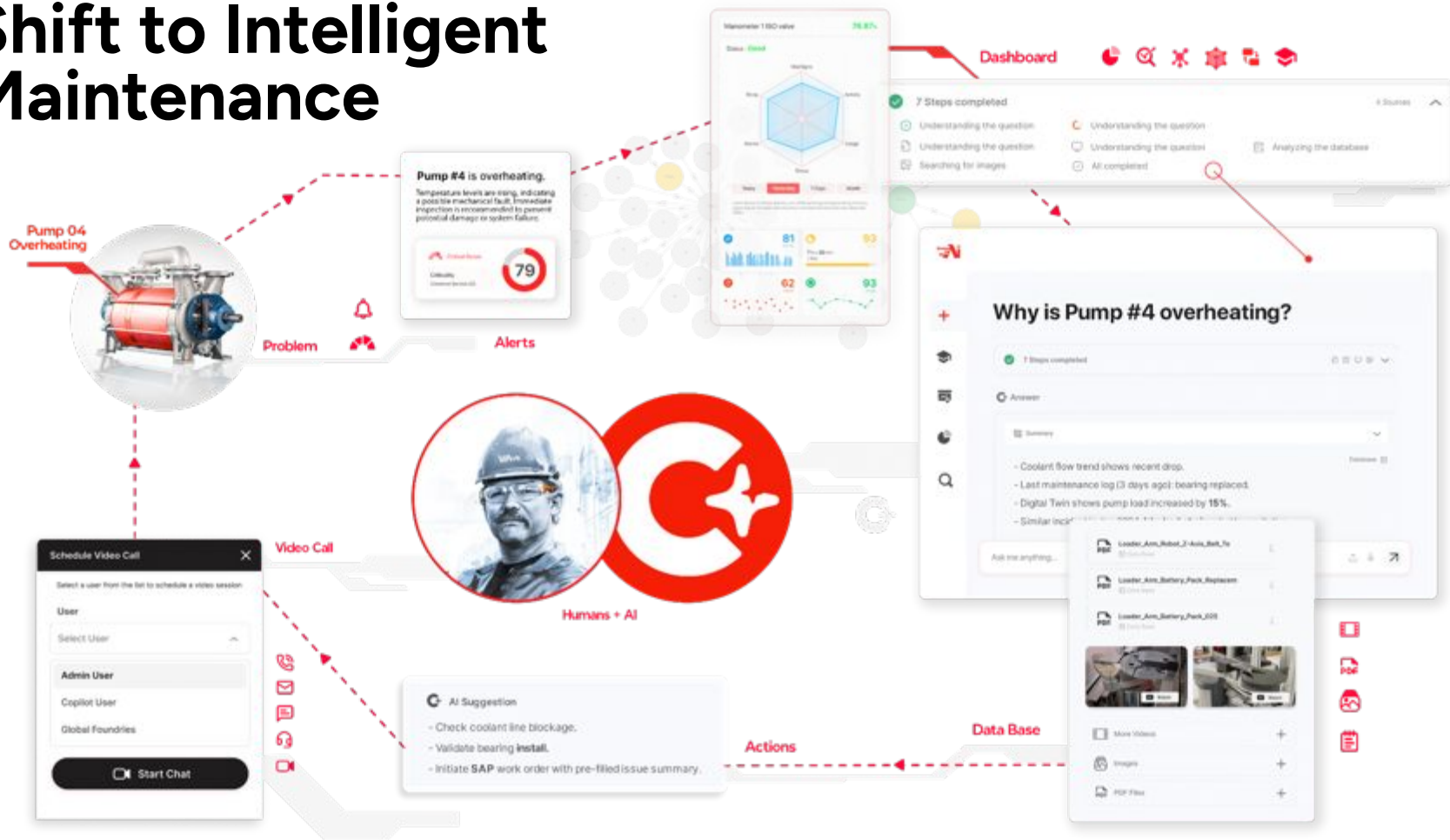
14% Attrition

High turnover disrupting operations and team consistency.

65% OEE

Poor maintenance can reduce equipment availability

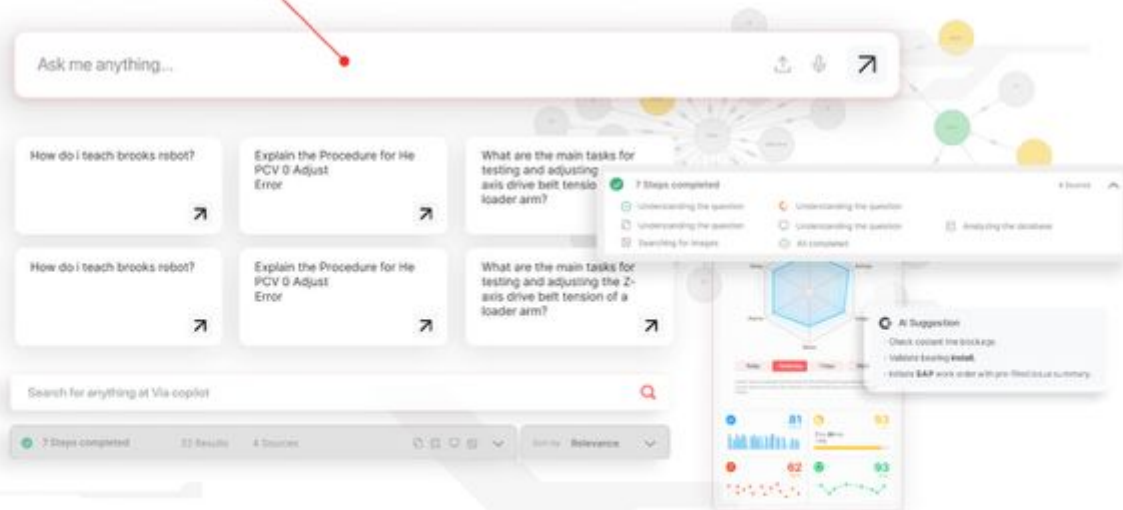
Shift to Intelligent Maintenance



Maintenance CoPilot



How Via Copilot
can help you today?



Multi-Agent Intelligence

RAG Agent Retrieves answers from manuals, SOPs, fault logs, and more.
Graph Agent Understands contextual relationships in equipment, documents, and processes through a dynamic Graph RAG.
DB Agent Pulls structured data from Oracle/MES.

Digital Twin Integration

Each asset is modeled digitally with a continuously updated health profile that tracks operating conditions, maintenance history, and sensor anomalies.

Real-Time & Historical Insight

Live sensor telemetry
(vibration, flow, temperature, pressure).
Historical data trends and past failure modes.
Work order history and parts data from SAP.
PLM insights like BoM, engineering change orders.

How is it different?

Digital Twin + Knowledge Graph

Via Automation integrates knowledge graphs into the digital twin framework. Knowledge graphs structure data for advanced queries and analytics, enhancing the capabilities of the digital twin. They map relationships between variables in industrial manufacturing. This allows for complex queries and relationship mapping, which are crucial for optimizing manufacturing processes.



Time Series & History

Analyze historical data to optimize performance and predict equipment behaviour.



Manuals & Documentation

Create and update guidelines for better equipment use and process control.



Sequence

AI detects defects, ensures quality, and maintains precision.



Knowledge Graph

Link data to boost decisions, diagnostics, and performance.



Visual Data

Model, optimize, predict, test cut downtime, boost efficiency.



Semantic Memory vs Episodic Memory

Semantic memory (stable “facts & rules”)

- In humans: vocabulary, concepts, generalized knowledge.
- In AI: the pre-trained & domain-fine-tuned weights of an LLM—recipes for oxide etch, drift-correction math, ISO 14644 clean-room classes, etc.

Episodic memory (time-stamped “experiences”)

- In humans: what happened at 10 a.m. yesterday.
- In AI: a knowledge graph (KG) whose nodes capture discrete FAB events—lot start/end, SPC violations, RF-generator drift at Tool ID DNE04 on 2025-06-10 14:37, the engineer who cleared the alarm, MTTR, downstream yield impact.

Fusion → LLM receives the retrieved event sub-graph, synthesizes: VIA AI

User prompt → “Why is Chamber B’s etch rate drifting today?”

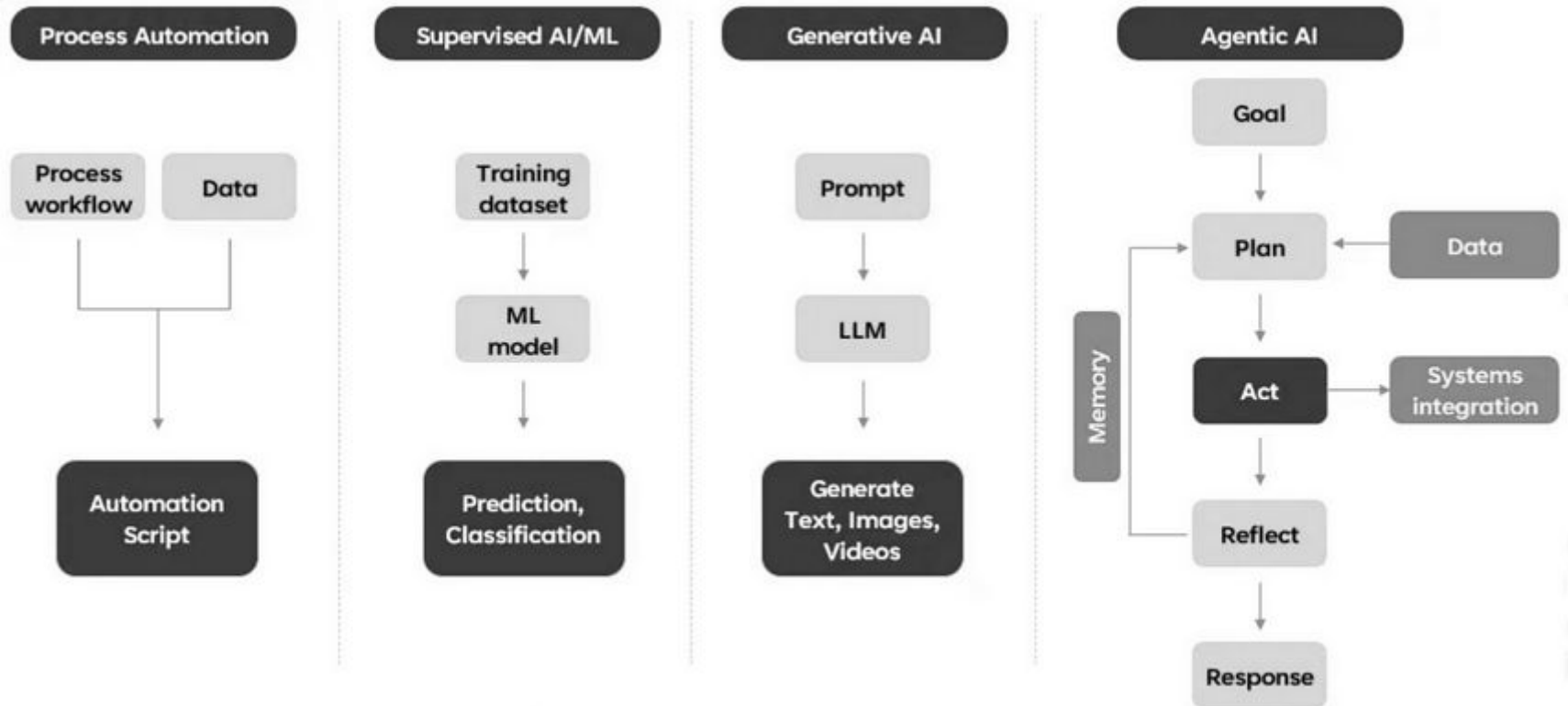
Semantic recall (LLM) → recalls drift causes (pressure, RF mismatch, polymer build-up) and suggests tests.

Episodic retrieval (KG) → pulls last 90 days of SECS/GEM data, PM logs, prior drift events, correlated OES spectra.

Fusion → LLM receives the retrieved event sub-graph, synthesizes:

- “Drift onset correlates (0.86 R) with helium leak alarms 48 h earlier.”
- “Mean MTTR has improved from 2.1 h to 1.4 h since the new kit install.”

Agentic AI Evolution



A Field Guide to AI Agents.

Fixed Automation – The Digital Assembly Line.

Level 0

Fixed Automation Agent.

Wafer Load/Unload Robot

Input Trigger: FOUP arrives at front-door sensor

Predefined Rule: If FOUP present, run pick-and-place sequence

Execute Action: Load wafer into chamber, close gate

Send Output: “Wafer loaded” signal to tool controller

Input Trigger

Predefined Rule

Execute Action

Send Output Result

LLM - Enhanced Agent.

Input Data

LLM (Contextual Analysis)

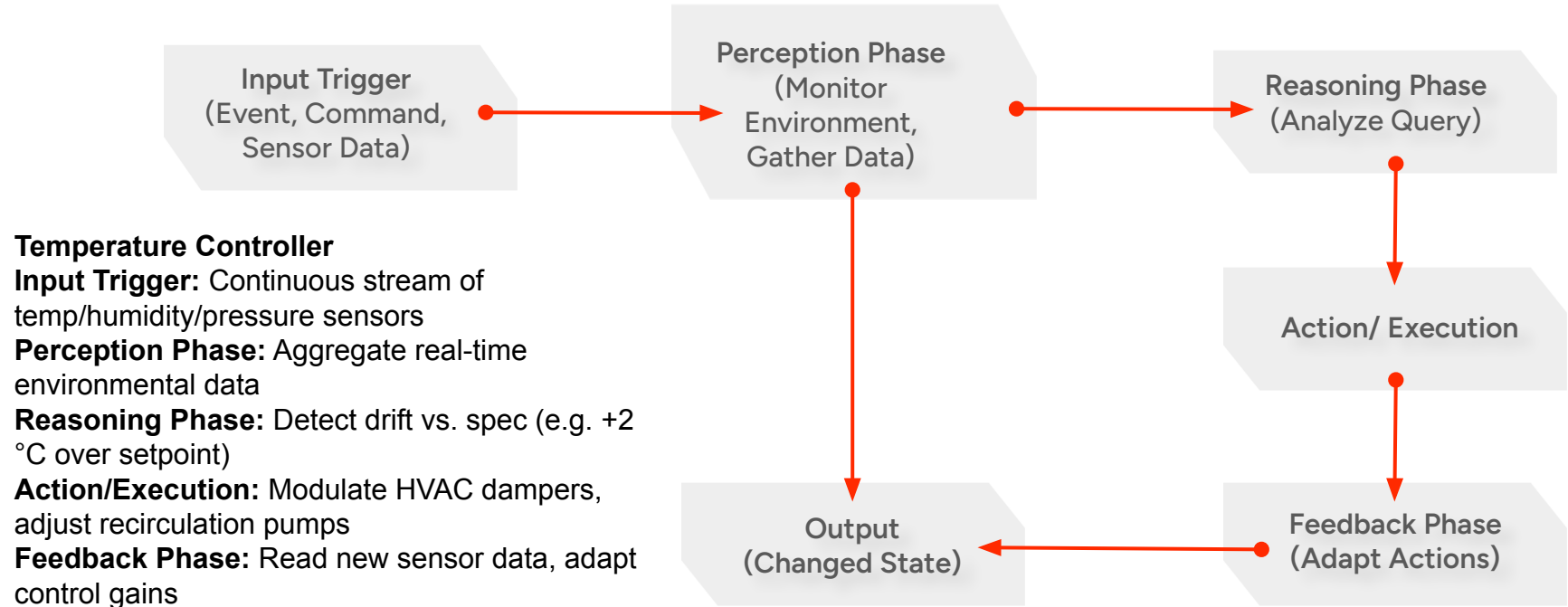
Rule-Based Constraint

Output/Action

A Field Guide to AI Agents.

Environment Controllers – The World Shapers.

Level 6

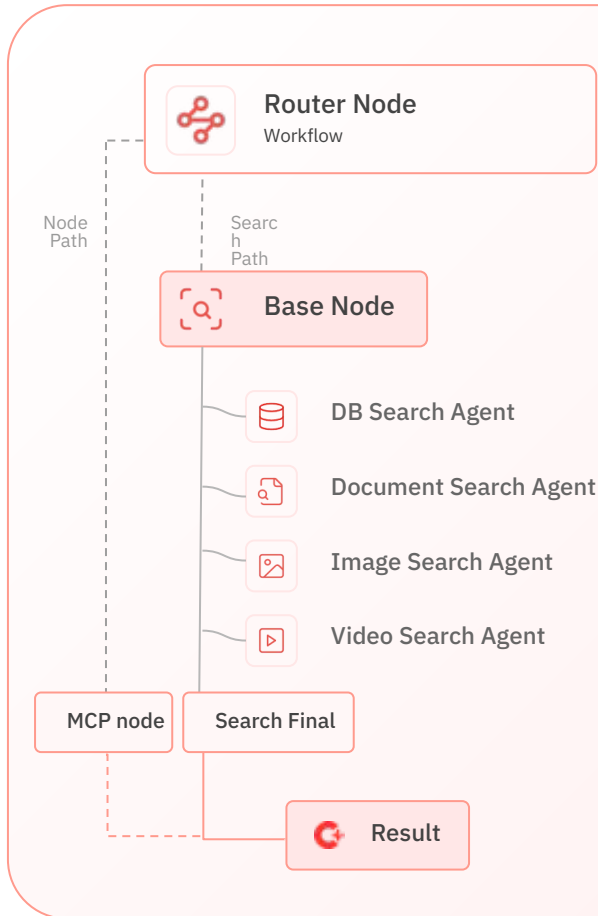


Maintenance Copilot Architecture – LLM + Agents.

- Modular agent system.
- Integrated with SCADA/MES/ERP.
- Real-time orchestration.
- Model Context Protocol routes intent.



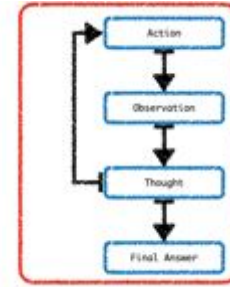
Via Agent Routing.



Agent | Workflow

Input

How do I fix Alarm with code 9127 ?



Output

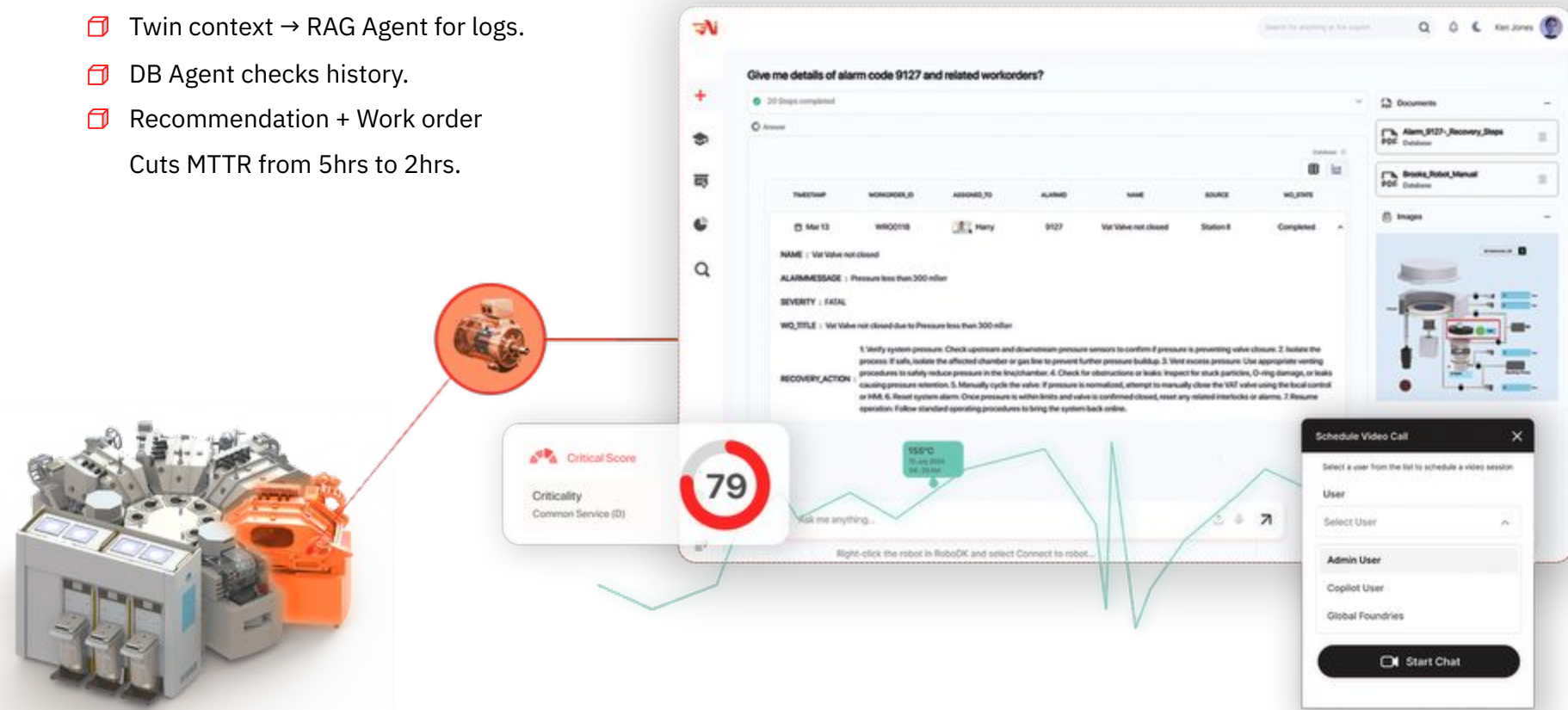
Alarm with code 9127 is because the VAT Valve did not close due to Pressure less than 300 mTorr. This alarm has occurred 5 times in the past 4 days. It was resolved by John Watson last time. Here are the recovery steps.

Recovery Actions:

- Verify system pressure: Check upstream and downstream pressure sensors to confirm if pressure is preventing valve closure.
- Isolate the process: If safe, isolate the affected chamber or gas line to prevent further pressure buildup.
- Vent excess pressure: Use appropriate venting procedures to safely reduce pressure in the line/chamber.
- Check for obstructions or leaks: Inspect for stuck particles, O-ring damage, or leaks causing pressure retention.

Case Study: Resolving Etch Tool Alarm (L5).

- Alarm → Graph Agent.
 - Twin context → RAG Agent for logs.
 - DB Agent checks history.
 - Recommendation + Work order
- Cuts MTTR from 5hrs to 2hrs.



Case Study: Resolving Process Variability(L7)

Problem

Incoming Wafer, Chamber variability, Tuning.

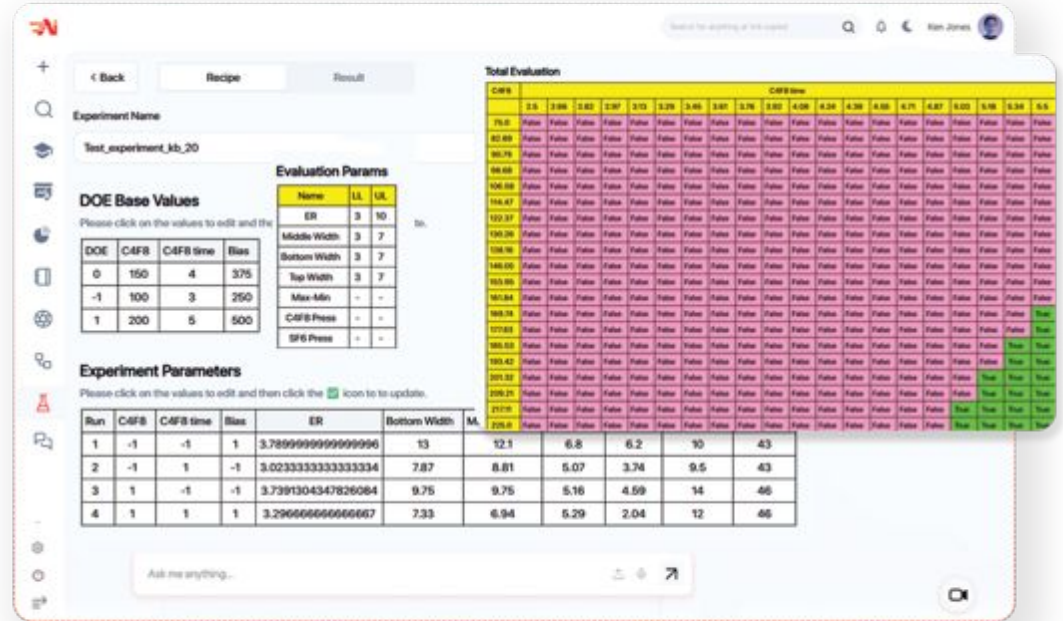
AI Agent Solution

- Experiment and Iterate.
- Virtual DOE with LLM + RAG Agent.
- Agent2Agent protocol iterates scenarios.

Results

Predictive Chamber Cleans

Faster packaging innovation cycles.





Industrial Workspace™

Unified Search. Instant Insight. FAB wide Intelligence.

Industrial Workspace™ is a unified enterprise search platform that connects all your factory data real-time, historical, structured, and Unstructured data into one intelligent interface.

Thank you.

